

Matplotlib Notes (Beginner + Intermediate)

1. Introduction to Matplotlib

Matplotlib is a Python library used for data visualization. It helps convert numerical data into graphs and charts. Common import: `import matplotlib.pyplot as plt`

2. First Plot

Use `plt.plot(x, y)` to create a line plot and `plt.show()` to display it.

3. Figure and Axes

Figure = entire canvas/window. Axes = graph area inside the figure. Professional code often uses: `fig, ax = plt.subplots()`.

4. Line Plots

Used to show trends. Important parameters: `color`, `linestyle`, `marker`, `linewidth`.

5. Labels and Titles

`xlabel()` labels the x-axis, `ylabel()` labels the y-axis, `title()` adds a chart title.

6. Legends

Used to identify multiple plotted series. Add labels in `plot()` and call `plt.legend()`.

7. Grid

`plt.grid()` adds guide lines that improve readability.

8. Axis Limits

`xlim()` and `ylim()` manually control displayed axis ranges.

9. Ticks

`xticks()` and `yticks()` customize displayed axis values and labels.

10. Saving Figures

`savefig()` saves graphs as PNG, JPG, PDF, etc. `dpi` controls image quality.

11. Scatter Plot

Shows individual points and is used to find relationships, correlations, and outliers. Use `plt.scatter()`. Parameter `s` controls size; `c` can specify multiple colors.

12. Bar Chart

Used to compare categories. Create with `plt.bar(categories, values)`.

13. Horizontal Bar Chart

Use `plt.barh()`. Best for long category names.

14. Histogram

Displays data distribution. Use `plt.hist(data)`. `bins` controls grouping ranges.

15. Pie Chart

Shows parts of a whole. Use `plt.pie()`. `labels` names slices; `autopct` displays percentages.

16. Subplots

Multiple graphs in one figure. Example: `plt.subplots(2,2)` creates four graph areas.

17. Figure Size

Control canvas size using `figsize=(width, height)`. Example: `figsize=(10,5)`.

18. Styles

Apply predefined themes using `plt.style.use()`. Examples: `ggplot`, `dark_background`, `default`.

19. Annotations

Add explanatory text to specific points using `plt.annotate()`.

20. Error Bars

Represent uncertainty or variability. Use `plt.errorbar()`. `yerr` specifies error values and `ecolor` sets error bar color.

21. Stem Plot

Used for discrete values. Create with `plt.stem(x, y)`.

22. Step Plot

Shows sudden level changes. Create with `plt.step(x, y)`.

Chart Selection Guide

Line Plot = trends, Scatter Plot = relationships, Bar Chart = category comparison, Histogram = distribution, Pie Chart = part-to-whole, Stem Plot = discrete values, Step Plot = sudden changes.